

Math 3A Podcast Problems
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Section 2.1 Tangent and Velocity Problems

1) The height of a rock thrown into the air is given by the function

$$h(t) = -16t^2 + 40t + 6.$$

a) Find the average velocity over the time interval $[1, 2]$.

b) Estimate the velocity as the rock hits the ground.

2) A particle moves a distance of $s(t) = \sin t + \cos 2t$ feet after t seconds. Find the average velocity of the particle over the first second of movement.

Section 2.2 Limits of Functions

3) Evaluate the limit $\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 - 4}$.

4) Evaluate the limit $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$.

Section 2.3 Calculating Limits

5) Evaluate the limit $\lim_{h \rightarrow 0} \left(\frac{1}{h} - \frac{1}{h^2 + h} \right)$.

6) Evaluate the limit $\lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h}$.

7) Evaluate the limit $\lim_{h \rightarrow 0} \frac{\frac{1}{5+h} - \frac{1}{5}}{h}$.

Section 2.5 Continuity

8) Explain why one function is continuous and the other is not.

$$f(x) = \begin{cases} \frac{x^2 - 6x + 9}{x - 3} & \text{if } x \neq 3 \\ 0 & \text{if } x = 3 \end{cases}$$

$$h(x) = \begin{cases} \frac{x^2 + 6x + 9}{x^2 - 9} & \text{if } x \neq 3 \\ 0 & \text{if } x = 3 \end{cases}$$

9) Determine if $f(x)$ is continuous $f(x) = \begin{cases} x + 1 & \text{if } x \leq 1 \\ \frac{1}{x} & \text{if } 1 < x < 3 \\ \sqrt{x - 3} & \text{if } x \geq 3 \end{cases}$

Section 2.6 Limits at Infinity

10) Evaluate the limit $\lim_{x \rightarrow \infty} \frac{2x + 9}{x^2 + 5x + 8}$

11) Evaluate the limit $\lim_{x \rightarrow \infty} \frac{10x^2 - 8}{x^2 + 2x + 1}$

12) Evaluate the limit $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 1})$

Section 2.7 Derivatives and Rates of Change

13) Find and graph the line that is tangent to $f(x) = \sqrt{x - 1}$ when $x = 2$.

14) The height of a baseball thrown straight up is given by the function $h(t) = -16t^2 + 50t + 6$.

a) Find the average velocity of the ball over the time interval $[1.9, 2]$.

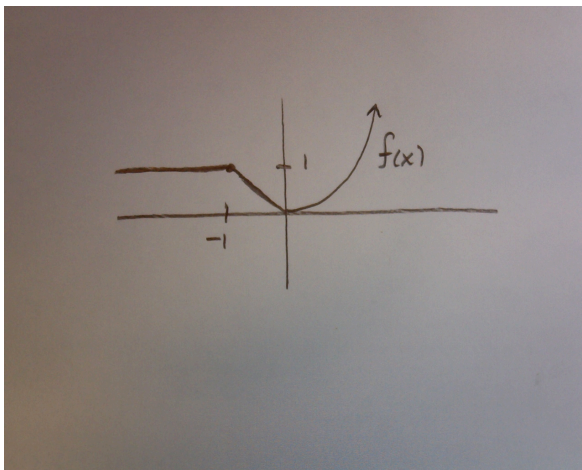
b) Find the instantaneous velocity of the ball at 2 seconds.

15) Draw the function $f(x)$ so that it has all of the following properties:

$$\begin{aligned} f(-1) &= f(1) = 0, f(0) = 1 \\ f'(-1) \text{ and } f'(0) &\text{ do not exist} \\ f'(x) &= -1 \text{ if } x < -1 \\ f'(x) &= 1 \text{ if } -1 < x < 0 \\ f'(x) &< 0 \text{ if } 0 < x < 1 \\ f'(x) &> 0 \text{ if } x > 1 \end{aligned}$$

Section 2.8 The Derivative

16) Given the graph of $f(x)$ below, sketch the graph of $f'(x)$.



17) The table below gives the percentage of Americans that are under the age 18 along with the years from 1950 to 2000.

t	$P(t)$
1950	31.1
1960	35.7
1970	34.0
1980	28.0
1990	25.7
2000	25.7

- What is the meaning of $P'(t)$? What are the units?
- Construct a table with estimates for $P'(t)$.
- Graph both $P(t)$ and $P'(t)$.

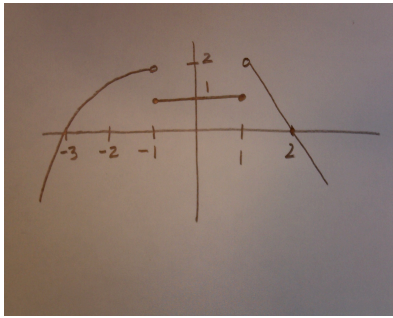
18) Use the definition to find $f'(a)$ for $f(x) = \sqrt[3]{x}$. Then explain why $f'(0)$ does not exist.

Chapter 2 Review

19) Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - x}{x^3 - 3x^2}$.

20) Use the definition to find $f'(3)$ for $f(x) = \frac{1}{x+5}$.

21) Given the graph of $f(x)$ below, answer a-d.



a) $\lim_{x \rightarrow -1} f(x) =$

b) $\lim_{x \rightarrow 1^-} f(x) =$

c) $f'(0) =$

d) $f'(-1) =$

Section 3.1 Derivatives of Polynomials and Exponential Functions

22) Given $f(x) = x^3 - 8x^2 + 2x + 3$, find all values of x where $f'(x) = 0$. Also, graph the function.

23) Find the first derivative of $y = \sqrt[3]{x}(x^2 - x)$.

24) Find the equation of the line tangent to $f(x) = x^4 + 2x^2 - x$ when $x = 1$.

Section 3.2 Product and Quotient Rules

25) Find the first derivative of $f(x) = \sqrt{x}e^x$.

26) Find the first derivative of $g(x) = \frac{1-x^2}{2x+3}$.

27) Use the product rule to prove the quotient rule.

Section 3.3 Derivatives of Trig Functions

28) Find a value in $[0, 2\pi)$ so that $\frac{dy}{dx} = 0$ for the function $y = 0.5x - \cos x$.

29) Find the equation of the line tangent to $y = x^2 \sin x$ when $x = 2$.

Section 3.4 The Chain Rule

30) Find all values in $[0, \pi)$ so that $\frac{dy}{dx} = 0$ for the function $y = \sin(x^2)$.

31) Find $f'(x)$ for the function $f(x) = \sqrt{\frac{x}{x^2 + 1}}$.

32) Find $f'(t)$ for the function $f(t) = \sin^2(\cos e^t)$.

Section 3.5 Implicit Differentiation

33) Find $\frac{dy}{dx}$ given that $y \tan x - x^3 y^3 = \sin y$.

34) Find $\frac{dy}{dx}$ for these three functions:

a) $x = \tan^{-1}(y^2)$

b) $x = \tan^{-1}(y)^2$

c) $x = \tan^2(y^{-1})$

Section 3.6 Derivatives of Logarithmic Functions

35) Find $f'(x)$ for the function $f(x) = \ln\left(\frac{1}{x}\right) + \frac{1}{\ln x} + \ln(\ln x)$.

36) Use logarithmic differentiation to find the first derivative of $y = (\cos x)^{\sin x}$.

Section 3.7 Rates of Change

37) The population of the US is modeled by the function $P(t) = 309674727e^{0.021t}$ where $t = 0$ represents July 1, 2010. Find and explain the meaning of $P'(10.5)$.

38) Volume of a sphere is given by $V = \frac{4}{3}\pi r^3$. Suppose that the radius of a spherical balloon is shrinking at a rate of $2 \text{ in} / \text{min}$. Find and explain what $\frac{dV}{dt}$ is when $r = 8 \text{ in}$.

Section 3.8 Exponential Growth and Decay

39) Let $A(t)$ be the amount of money in an account where \$1000 was invested and 8% interest is paid compounded continuously. Find and explain each of the following: $A(10)$, $A'(10)$ and $A''(10)$.

40) A bacteria culture grows with a constant relative growth rate. After 2 hours there are 800 bacterial and after 10 hours there are 80,000 bacteria. Find an expression for the size of the population after t hours.

Section 3.9 Related Rates

41) An object is travelling on the path $x^2 + y^2 = 9$ with $\frac{dx}{dt} = -10 \text{ m} / \text{s}$ when $x = 2$.

Find $\frac{dy}{dt}$ and explain what it has to do with the object.

42) Gravel is being dumped from a conveyor belt at a rate of $40 \text{ ft}^3 / \text{min}$. The coarseness of the gravel is such that the pile formed is a cone whose base diameter and height are always the same. How fast is the height of the pile increasing when the pile is 10 feet tall?

Section 3.10 Differentials

43) Compute dy and Δy given that $\Delta x = 1$ and $x = 1$ for the function $y = \sqrt{x}$.

44) The volume of a cube is given to be 80 ft^3 with a maximum error in side measurement to be 0.01 feet. Use differentials to estimate the possible error in the volume.

Chapter 3 Review

45) Find $\frac{dy}{dx}$ for each of the following functions:

a) $y = \tan(\tan x)$ b) $y = \tan x(\tan x)$ c) $y = e^{(\tan x)}$ d) $y = \tan(e^x)$

46) Find the equation of the line tangent to $x^2 + 4xy + y^2 = 13$ at the point $(2,1)$.

47) Find any point where $\frac{dg}{dx} = 0$ for the function $g(x) = \tan^{-1}(\sin^{-1} \sqrt{x})$.

Section 4.1 Maximum and Minimum Values

48) Find all minimum and maximum values for $f(x) = x^3 - 5x + 6$.

49) Find all minimum and maximum values for $f(x) = x - \ln x$.

50) Find all minimum and maximum values in the interval $[0, 2\pi)$ for $y = \cos(2x) + \sin x$.

Section 4.2 The Mean Value Theorem

51) Show that the equation $2x - 1 = \sin x$ has exactly one real root.

52) At 2:00PM a car's speedometer read 30 mi/h and at 2:10PM it read 50 mi/h . Prove that at some time between 2:00 and 2:10, the car was accelerating at 120 mi/h^2 .

Section 4.3 Derivatives and Shapes of Graphs

53) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values.

$$f(x) = \ln(x^4 + 8)$$

54) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values.

$$y = \sin x - \cos x \text{ on the interval } [0, 2\pi)$$

55) Sketch the graph of the continuous function $f(x)$ with all of the following properties:

$$f(-3) = f(0) = f(3) = 0$$

$$f'(-2) = 0, f'(2) \text{ DNE}, f'(x) > 0 \text{ on the interval } (-2, 2)$$

$$f''(x) > 0 \text{ on } (-3, 2), f''(x) = 0 \text{ on } [2, \infty)$$

Section 4.4 L'Hospital's Rule

56) Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x}$.

57) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2}}{\sqrt{2x^2 + 1}}$.

58) Evaluate $\lim_{x \rightarrow -\infty} x^2 e^x$.

Section 4.5 Curve Sketching

59) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = \frac{x^2}{x^2 + 4}$$

60) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = x + \sin x$$

61) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = \frac{\ln x}{x}$$

Section 4.7 Optimization Problems

62) *Designing a poster.* You are designing a rectangular poster to contain 50 in² of printing with a 4-inch margin at the top and bottom and a 2 inch margin at each side. What are the dimensions that will minimize the amount of paper used?

63) The U.S. Postal Service will accept a box for domestic shipment only if the sum of the length and girth (distance around, like perimeter) does not exceed 108 inches. What dimensions will give a box with a square end the largest possible volume?

64) *Shortest Beam.* There is an 8-foot tall wall that stands 27 feet away from a building. Find the length of the shortest straight beam that will reach to the side of the building from the ground outside the wall.

Section 4.8 Newton's Method

65) Use Newton's Method to estimate the only solution for $e^x = -2x + 3$.

66) Use Newton's Method to estimate the smallest positive solution for $\sin x = x^2$.

Section 4.9 Antiderivatives

67) Find the antiderivative for $g(x) = \frac{1}{x^2} + \frac{1}{x} + \sec^2 x$.

68) Find $f(x)$ if $f'(x) = \sqrt[3]{x} + 6x^4$ and $f(0) = 2$.

69) Find $f(x)$ if $f''(x) = \sin x + \cos x$ and $f'(0) = 1, f(0) = 2$.

Section 5.1 Areas and Distances

70) Consider the region that is above the x -axis and the function $y = -x^2 + x + 6$. Estimate this area as accurately as possible using 5 rectangles.

71) Estimate the total area between $y = \sin x$ on $[0, \pi]$ and the x -axis using 10 rectangles.

Section 5.2 The Definite Integral

72) Use the Midpoint Rule to estimate $\int_0^3 (x^2 - 3x) dx$ using 12 rectangles.

73) It is a fact that $\int_0^{\pi/2} \cos x \, dx = 1$. Find the least number of rectangles required to estimate the definite integral within 0.001 of the actual answer.

Section 5.3 The Fundamental Theorem of Calculus

74) Evaluate $\int_0^x (t^4 - 3t) \, dt$ and notice that the result is a function of x . Take the derivative of your answer. How does the result compare to the original problem?

75) Find the area between the x -axis and the function $f(x) = \frac{2}{x^2 + 1}$ over the interval $[0, 1]$.

Section 5.4 Indefinite Integrals

76) Evaluate $\int \left(\frac{3}{\sqrt{x}} + x^2 + \frac{1}{x} \right) dx$.

77) Evaluate $\int \sec x (\sec x + \tan x) \, dx$.

78) Evaluate $\int \frac{\sin 2x}{\sin x} \, dx$.

Section 5.5 The Substitution Rule

79) Evaluate $\int (x + 3)\sqrt{x^2 + 6x + 1} \, dx$.

80) Evaluate $\int \frac{x}{\sqrt{1 - x^2}} \, dx$ and $\int \frac{x}{\sqrt{1 - x^4}} \, dx$.

81) Evaluate $\int x^3 \sqrt{x^2 + 3} \, dx$.

Chapter 5 Review

82) Use Riemann sums with $n = 50$ to estimate $\int_0^4 \frac{x}{1+x^2} dx$. Next use integration to calculate the exact answer.

83) Find the derivative of $g(x) = \int_1^{x^3} \frac{e^t}{t} dt$.

84) Evaluate $\int_0^1 (\sqrt[4]{x} + 1)^2 dx$

Section 6.1 Area Between Curves

85) Find the area in QI that is bound by the given curves $y = 3x^2$, $y = 8x^2$, $4x + y = 4$.

86) Find the total area between $y = \sin 2x$ and $y = \cos x$ over the interval $[0, \pi]$.

Section 6.2 Volumes

87) Consider the finite region in QI that is bound by $y = x^2$ and $y = x^4$. Find the volume if that region were rotated about the x -axis.

88) Find the volume created when the region defined by $y = 1 + \sec x$ and $x = 0$ on the interval $[0, 1]$ is rotated about the x -axis.

Section 6.3 Volumes by Cylindrical Shells

89) Consider the finite region in QI defined by $y = -x^2 + 2x$ and $x = 0$. Find the volume if the region was rotated about the y -axis.

90) Consider the finite region in QI defined by $y = -x^2 + 2x$ and $x = 0$. Find the volume if the region was rotated about the line $x = 2$.

91) Consider the finite region in QI defined by $y = -x^2 + 2x$ and $x = 0$. Find the volume if the region was rotated about the line $y = 1$.

Section 6.4 Work

92) A cable that weighs 3 pounds per foot is used to lift 500 pounds of coal up a 400 foot mine shaft. How much work was done?

93) A cylindrical tank with a radius of 3 feet and a height of 8 feet is full of water. Find the amount of work done if all of the water is pumped out of the top of the tank.

Section 6.5 Average Value of a Function

94) If a baseball were thrown with an initial velocity of $40 \frac{\text{ft}}{\text{sec}}$ and an initial height of 6 ft, then the height after t seconds would be $h(t) = -16t^2 + 40t + 6$. Use this to find the average velocity of the baseball during its flight through the air.

95) Find the average value of $y = \sin(2x)\cos^2 x$ over the interval $[0, \frac{\pi}{2}]$.

Final Review

96) Find $\frac{dy}{dx}$ a) $e^y = \ln x$ b) $y = \sin^2 x$ c) $x = \tan y$ d) $y = \sin^{-1}(5x)$

97) a) $\int \frac{x^2}{x^3 + 3} dx$ b) $\int \frac{2x+1}{3x-5} dx$ c) $\int \cos^2 x dx$

98) Use the definition of the derivative to prove: $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$.

99) Use derivatives to describe the graph of $f(x) = \cos^2 x$ on the interval $[0, 2\pi]$.

100) A water tank is the shape of a right circular cylinder of radius 7 feet. Water is leaking out of the tank at a rate of $0.5 \frac{\text{ft}^3}{\text{min}}$. Find the rate at which the water level is falling after 13 minutes.